

LESSON 5

INDUCTIVE AND DEDUCTIVE REASONING

Overview:

This lesson will expose students to different types of reasoning and how these reasoning types may be used in solving problems in Mathematics and even real life problems. Solving puzzles will also be incorporated in this lesson to add fun and excitement.

Learning Outcomes:

At the end of lesson 5, students can:

1. determine arguments that shows inductive reasoning or deductive reasoning;
2. differentiate inductive reasoning from deductive reasoning;
3. distinguish whether the statement is correct using inductive reasoning; and
4. apply inductive reasoning and deductive reasoning in solving application problems.

Materials Needed:

- illustrations
- puzzles
- supplementary video
- learning module

Duration: 3 hours

Learning Content:

5.1 INDUCTIVE REASONING

Inductive Reasoning is a process of reaching a general conclusion by examining specific examples.

Inductive reasoning has many applications in solving problems. It can be used to predict the next number in the list according to some patterns you have observed, make conjecture about an arithmetic procedure, etc.

Example

1. Use the inductive reasoning to predict the next number in each of the following lists.
- a) 3, 6, 9, 12, 15, ? b) 1, 3, 6, 10, 15, ?

Solution

- Each successive number is larger than the preceding number. Thus we predict the next number in the list is 3 larger than 15, which is 18.
- The first two numbers differ by 2. The second and the third numbers differ by 3. It appears that the difference between any two numbers is always 1 more than the preceding difference. Since 10 and 15 differ by 5, we predict that the next number in the list will be 6 larger than 15, which is 21.

Example

2. Consider the following procedure: pick a number. Multiply the number by 8, add 6 to the product, divide the sum by 2, and subtract 3.
- Complete the above procedure for several different numbers. Use inductive reasoning to make a conjecture about the relationship between the size of the resulting number and the size of the original number.

Solution

Suppose we pick 5 as our original number. Then the procedure would produce the following results:

Original Number:	5	
Multiply by 8:	8×5	$= 40$
Add 6:	$40 + 6$	$= 46$
Divide by 2:	$46 \div 2$	$= 23$
Subtract 3:	$23 - 3$	$= 20$

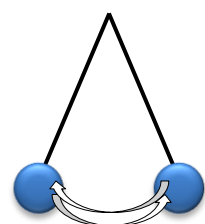
We started with 5 and followed the procedure to produce 20. Starting with 6 as our original number produces a final result of 24. Starting with 10 produces a final result of 40. Starting with 100 produces a final result of 400. In each of these cases the resulting number is four times the original number. We conjecture that following the given procedure produces a number that is four times the original number.

Scientists often use inductive reasoning. For instance, Galileo Galilei (1564-1642) used inductive reasoning to discover that the time required for a pendulum

to complete one swing, called the period of the pendulum, depends on the length of the pendulum. Galileo did not have a clock, so he measured the periods of pendulums in “heartbeats.”

The following table shows some results obtained for pendulums of various lengths. For the sake of convenience, a length of 10 inches has been designated as 1 unit.

Length of pendulum, in units	Period of pendulum, in heartbeats
1	1
4	2
9	3
16	4
25	5
36	6



The period of a pendulum is the time it takes for the pendulum to swing from left to right and back to its original position.

Example

- 3. Use the data in the above table and inductive reasoning to answer each of the following questions.
 - a. If a pendulum has length of 49 units, what is its period?
 - b. If the length of a pendulum is quadrupled, what happens to its period?

Solution

- a) In the table, each pendulum has a period that is the square root of its length. Thus we conjecture that a pendulum with a length of 49 units will have a period of 7 heartbeats.
- b) In the table, a pendulum with a length of 4 units has a period that is twice that of a pendulum with a length of 1 unit. A pendulum with a length of 16 units has a period that is twice that of a pendulum with a length of 4 units. It appears that quadrupling the length of a pendulum doubles its periods.

5.2 DEDUCTIVE REASONING

Deductive Reasoning is the process of reaching a conclusion by applying general assumptions, procedures, or principles.

Deductive reasoning can be used solving a logic puzzle as shown in the following examples.

Example

- 1. Use deductive reasoning to show that the following procedure produces a number that is four times the original number.

Procedure: Pick a number. Multiply the number by 8, add 6 to the product, divide the sum by 2, and subtract 3.

Solution

Let n represent the original number.

Multiply the number by 8:
$$8n$$

Add 6 to the product:
$$8n + 6$$

Divide the sum by 2:
$$\frac{8n+6}{2} = 4n + 3$$

Subtract 3:
$$4n + 3 - 3 = 4n$$

We started with n and ended with $4n$. The procedure given in this example produces a number that is four times the original number.

Example

2. Each of four neighbors, Sean, Maria, Sarah, and Brian, has a different occupation (editor, banker, chef, or dentist). From the following clues below, determine the occupation of each neighbor.
- a) Maria gets home from work after the banker but before the dentist.

b) Sarah, who is the last to get home from work, not the editor.

c) The dentist and Sarah leave for work at the same time.

d) The banker lives next door to Brian.

Solution

From clue a, Maria is not the banker or the dentist. In the following chart, write $\times_a$ (which stands for “ruled out by clue a”) in the Banker and the Dentist columns of Maria’s row.

	Editor	Banker	Chef	Dentist
Sean				
Maria		$\times_a$		$\times_a$
Sarah				
Brian				

From clue b, Sarah is not the dentist. Write $\times_b$ (ruled out by clue b) in the Editor column of Sarah’s row. We know from clue a that the banker is not the last to get home, and we now from clue b that Sarah is the last to get home; therefore, Sarah is not the banker. Write $\times_b$ in the Banker column of Sarah’s row.

	Editor	Banker	Chef	Dentist
Sean				
Maria		Xa		Xa
Sarah	Xb	Xb		
Brian				

From clue c, Sarah is not the dentist. Write $\times_c$ for this condition. There are now Xs for three of the four occupations in Sarah's row; therefore, Sarah must be the chef. Place a $\checkmark$ in that box. Since Sarah is the chef, none of the other 3 people can be the chef. Write $\times_c$ for these conditions. There are now Xs for three of the four occupations in Maria's row; therefore, Maria must be the editor. Insert a $\checkmark$ to indicate that Maria is the editor, and write $\times_c$ twice to indicate that neither Sean nor Brian is the editor.

	Editor	Banker	Chef	Dentist
Sean	Xc		Xc	
Maria	$\checkmark$	Xa	Xc	Xa
Sarah	Xb	Xb	$\checkmark$	Xc
Brian	Xc		Xc	

From clue d, Brian is not the baker. Write Xd for this condition. See the following table. Since there are three Xs in the Banker column, Sean must be the banker. Place

in that box. Thus Sean cannot be the dentist. Write $\times_d$ in that box. Since there are 3 Xs in the Dentist column, Brian must be the dentist. Place a $\checkmark$ in that box.

	Editor	Banker	Chef	Dentist
Sean	Xc	$\checkmark$	Xc	Xd
Maria	$\checkmark$	Xa	Xc	Xa
Sarah	Xb	Xb	$\checkmark$	Xc
Brian	Xc	Xd	Xc	$\checkmark$

Sean is the banker, Maria is the editor, Sarah is the chef, and Brian is the dentist.

Learning Activity:

Determine whether each of the following arguments is an example of inductive reasoning or deductive reasoning. Explain your answer.

1. During the past 10 years, a tree has produced plums every other year. Last year the tree did not produce plums, so this year the tree will produce plums.
2. All home improvements cost more than the estimate. The contractor estimated that my home improvement will cost ₱350,000. Thus my home improvement will cost more than ₱350,000.
3. All Gillian Flynn novels are worth reading. The novel *Gone Girl* is a Gillian Flynn novel. Thus *Gone Girl* is worth reading.
4. I know I will win a jackpot on this slot machine in the next 10 tries, because it has not paid out any money during the last 45 tries.

Learning Evaluation:

- A. Use the inductive reasoning to predict the next number in each of the following lists.
1. 4, 8, 12, 16, 20, 24, ?
 2. 80, 70, 61, 53, 46, 40, ?
 3. $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, ?$
 4. 2, 7, -3, 2, -8, -3, -13, -8, -18, ?
- B. Use the inductive reasoning to decide whether each statement is correct or not.
1. The sum of any two even counting numbers is always an even counting numbers.
 2. Pick any counting number. Multiply the number by. Add 8 to the product. Divide the sum by 2. Subtract 4 from the quotient. The resulting number is twice the original number.
- C. Use deductive reasoning to show that the following procedure always produces a number that is equal to the original number.
- Procedure:* Pick a number. Multiply the number by 6 and add 8. Divide the sum by 2, subtract twice the original number, and subtract 4.

D. Use deductive reasoning to determine the missing number in each magic square.

(A **magic square of order n** is an arrangement of numbers in a square such that the sum of the n numbers in each row, column, and diagonal is the same number.)

Example of magic square of order 3:

4	9	2
3	5	7
8	1	6

1. A magic square of order 4

		2	13
	10	11	
	6		12
4	15		1

2. A magic square of order 5

11		7		3
4			8	16
	5	13		
10	18	1		22
23	6		2	15

References:

Aufman, R. et. al. (2018). *Mathematics in the Modern World*. Rex Book Store Inc. Sampaloc, Manila, Philippines.

Baningon Jr.,R. et.Al. (2016). *Statistics and Probability*. Educational Resources Corporation. Cubao, Quezon City.